

Module 1: AI and Risk - Introduction and Overview

100 Tutorial Topics: Classical AI, Neural Networks, Machine Learning Types, AI Risks

GARP RAI Exam Preparation

TOPIC: What is Classical AI?

DEFINITION:

- The original paradigm of AI research
- Based on explicit, precise processing
- Typified by symbolic logic and formal rule-governed systems
- Also known as "Good Old-Fashioned AI" (GOFAI)

HISTORICAL ROOTS:

- Charles Babbage (early 19th century): Difference engine and analytical engine
- Alan Turing (1936): Theoretical "Turing machine"
- Turing's 1950 paper "Computing Machinery and Intelligence"
- The "imitation game" (Turing Test)

KEY CHARACTERISTICS:

- Intelligence as explicit rule-following
- Symbolic representation
- Logical deduction
- Search through possibilities

EXAM TRAP: Classical AI = explicit, rule-based processing!

T2: Turing Test

TOPIC: The Turing Test (Imitation Game)

DEFINITION:

- Proposed by Alan Turing in his 1950 paper
- Originally called the "imitation game"
- A test of machine intelligence

THE TEST:

- A computer proves itself intelligent if it can generate textual conversation
- Quality indistinguishable from an intelligent human
- Ranging over any topic chosen by the human "interrogator"

SIGNIFICANCE:

- Thought experiment demonstrating possibility of AI
- Highly controversial philosophically
- Remains influential
- Particularly salient with modern chatbots (ChatGPT)

LIMITATIONS:

- Focuses on conversational ability, not general intelligence
- Can be gamed by sophisticated mimicry
- Doesn't require understanding, only imitation

EXAM TRAP: Turing Test = [indistinguishable conversation](#), not true understanding!

TOPIC: Specific versus General AI

SPECIFIC (NARROW) AI:

- Designed to be intelligent at solving specific problems
- Outperforms humans in specific domains
- Examples: calculating tables, generating statistics, playing chess
- Most practical AI applications

GENERAL AI (AGI):

- Comparable to human general intelligence
- Can handle any intellectual task
- Source of existential risk concerns
- Currently theoretical

KEY INSIGHT:

- A system can be "intelligent" in specific tasks without general competence
- Turing's test was a conceptual existence proof
- Practical AI focuses on specific problem-solving

EXAM TRAP: Most AI is **narrow/specific**, not general!

TOPIC: The Dartmouth Conference (1956)

KEY FACTS:

- Term "artificial intelligence" coined by John McCarthy
- Organized by McCarthy, Minsky, Rochester, and Shannon
- Two-month summer conference at Dartmouth College
- Seminal event in AI history

CORE CONJECTURE:

- "Every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it"
- Intelligence as explicit, precise processing
- Symbolic logic as paradigm

SIGNIFICANCE:

- Established AI as a research field
- Set the agenda for decades
- Echoed Turing's notion of universality

EXAM TRAP: Dartmouth Conference (1956) = birth of AI as a field!

TOPIC: Logic Theorist Program

KEY FACTS:

- Developed by Newell, Simon, and Shaw (1955)
- Designed to generate proofs in propositional logic
- Presented at Dartmouth conference
- Confirmed most theorems in Principia Mathematica

ACHIEVEMENTS:

- Found new and more elegant proof of Theorem 2.85
- Demonstrated machine intelligence in logic
- High-profile application of AI

SIGNIFICANCE:

- Early success of classical AI
- Validated logicist thesis (mathematics from logic)
- Demonstrated rule-based reasoning

EXAM TRAP: Logic Theorist = automated theorem proving!

TOPIC: Why Games Were Attractive AI Testbeds

THREE KEY REASONS:

1 Relatively Simple:

- Compared to complexities of normal life
- Could be tackled with limited computing resources
- No need to record physical environment details

2 Straightforwardly Rule-Governed:

- Clear and explicit regulations
- Setup of the game
- How pieces are permitted to move
- Effects of moves on board position
- Criteria for winning, losing, drawing

3 Competitive:

- Players can be rated by relative skill
- Natural assessment of progress
- Straightforward to judge improvement

EXAM TRAP: Games provided **controlled environments** for AI development!

T7: Reinforcement Learning - Nim Example

TOPIC: Simple Reinforcement Learning - Nim

THE GAME OF NIM:

- 20 coins on a table
- Players alternate removing 1, 2, or 3 coins
- Winner removes the last coin(s)
- Winning strategy: reduce to multiple of 4

LEARNING MECHANISM:

- Assign weights to each move in each position
- Initially all weights equal (e.g., 5)
- Choose moves randomly based on weights
- After each game, adjust weights based on outcome

REINFORCEMENT RULES:

- **Winner:** Chosen moves that led to victory are positively reinforced (weight +1)
- **Winner:** Moves not chosen are negatively reinforced (weight -1)
- **Loser:** Chosen moves that led to loss are negatively reinforced
- **Loser:** Moves not chosen are positively reinforced

RESULT:

- After 100 games, discovers winning strategy
- Learns to reduce coins to multiples of 4
- No explicit "lookahead" or planning

EXAM TRAP: Reinforcement learning = learning from trial and error!

TOPIC: Lookahead in AI

DEFINITION:

- Forming a multistep plan to achieve a goal
- Exploring consequences of possible actions
- Essential when many possible situations exist

WHEN LOOKAHEAD IS ESSENTIAL:

- Very large number of possible situations overall
- Relatively constrained and predictable options in particular situations
- Need to form multistep plans

EXAMPLES:

- **Sudoku**: "If I put 4 here, then this cell must be 7..."
- **Tower of Hanoi**: Strategic planning to move disks
- **Chess**: Considering future moves and responses

APPROACHES:

- Tactical lookahead: Narrow down plausible sequences
- Strategic planning: Identify useful shorter-term outcomes

EXAM TRAP: Lookahead = exploring future consequences of actions!

TOPIC: Search in Classical AI

DEFINITION:

- Generating sets of possible outcomes
- Searching through them for best achievement of aims
- Fundamental technique in AI

BINARY SEARCH:

- Progressively divide search space in two
- Used in number-guessing program
- Finds target within at most 10 guesses for 1,000 possibilities
- Efficient: $2^{10} = 1,024$

ROUTE-FINDING SEARCH:

- **Breadth-First Search:** Track all possible routes, build stage by stage
- **Depth-First Search:** Follow each route to end before comparing next
- **Heuristic Search:** Use additional information to guide exploration

A* ALGORITHM:

- Best-known heuristic search technique
- Tracks distance traveled so far
- Estimates distance still to go (optimistic)
- Orders locations by sum of these values

EXAM TRAP: Search = systematic exploration of possibilities!

TOPIC: Combinatorial Explosion

DEFINITION:

- Search space grows enormously as problem size increases
- Relevant numbers of possible choices get multiplied together
- Certainty of solution quickly becomes unfeasible

EXAMPLES:

- **Travelling Salesman Problem:** Finding optimal route around cities
- **Chess:** Average branching factor 30
- **Go:** Average branching factor 250
- **Checkers:** Average branching factor 10

CONSEQUENCES:

- Exhaustive search impractical for complex problems
- Earth would be swallowed by Sun before first move
- Need for heuristics and approximations
- Adversarial search adds further complications

EXAM TRAP: Combinatorial explosion = exponential growth in possibilities!

TOPIC: Recursion in AI

DEFINITION:

- Powerful technique in classical AI
- Complex problem solved by reducing to simpler instances of same problem
- Huge importance in computer science generally

TOWER OF HANOI EXAMPLE:

- Move pile of disks from pillar A to pillar B
- Break down into three subtasks:
 - 1 Move pile of $n-1$ disks from A to C
 - 2 Move disk n from A to B
 - 3 Move pile of $n-1$ disks from C to B
- Each subtask reduces to simpler instance
- Eventually reduces to single disk movements

MATHEMATICAL RESULT:

- Moving n disks takes $2^n - 1$ steps
- Moving 9 disks takes 511 steps
- Proved by mathematical induction

EXAM TRAP: Recursion = solving problems by reducing to simpler versions!

T12: Recursive Adversarial Tree Search

TOPIC: Recursive Adversarial Tree Search - Noughts and Crosses

GOAL:

- Design program to play perfect game of Noughts-and-Crosses
- Never lose
- Always find winning sequence if one exists

POSITION VALUES:

- **+1**: Win for player whose turn it is
- **-1**: Loss for player whose turn it is
- **0**: Draw

ALGORITHM:

- 1 If X has lost (opponent has line of three), value = -1
- 2 If position is full, value = 0 (draw)
- 3 Otherwise, construct all positions with Y to move
- 4 Evaluate each from Y's point of view using same algorithm
- 5 Assign X-value = inverse of lowest Y-value

MINIMAX SEARCH:

- Presume opponent chooses reply maximizing their benefit
- Choose move minimizing that maximum benefit
- Applied recursively at every level

EXAM TRAP: Adversarial search = **minimax principle**!

T13: Heuristics and Refinement

TOPIC: Heuristics and Refinement by Reinforcement Learning

HEURISTICS DEFINITION:

- Calculable criteria or rules of thumb
- Guide choice of position to aim for
- Cannot guarantee optimal solution

CHECKERS EXAMPLE - CRITERIA:

- 1 Material balance (my pieces vs. opponent's)
- 2 Relative mobility of pieces
- 3 How far advanced pieces are
- 4 Relative number of "kings"
- 5 Additional criteria (Samuel devised 38)

WEIGHTED EVALUATION:

$$v = w_1 c_1 + w_2 c_2 + w_3 c_3 + w_4 c_4 + w_5 c_5 + w_6 c_6$$

REFINEMENT BY REINFORCEMENT LEARNING:

- Initial weights equal (e.g., 1,000)
- Random variation to create competing strategy
- Play games between current and competing
- Replace current with competing if better
- Repeat iteratively

EXAM TRAP: Heuristics guide search; **weights refined by reinforcement learning!**

TOPIC: Limitations of Classical AI

KEY DIFFICULTIES:

1 Representation Problem:

- How to represent characteristics and relationships
- Especially rules governing behavior
- Experts find it hard to elucidate implicit assumptions

2 Frame Problem:

- Tracking what changes when actions performed
- Also tracking what stays the same
- Adds to combinatorial explosion

3 Uncertainty:

- Ambiguous or uncertain data
- Complex probability calculations
- Unlike perfect-information games

4 Real-Time Data:

- Error-prone acquisition
- Computer vision challenges
- Object recognition difficulties

EXAM TRAP: Classical AI is **brittle** - fails beyond narrow boundaries!

TOPIC: Expert Systems

DEFINITION:

- Designed to capture knowledge within specific domains
- Often use logic programming
- Response to limitations of general-purpose AI
- Especially from 1980s

ARCHITECTURE:

- **Knowledge Base:** Topic-specific facts and rules
- **Inference Engine:** General-purpose reasoning mechanism
- Knowledge engineers extend to new domains

EXAMPLE - JAPANESE FIFTH GENERATION:

- Massive government investment
- Harness PROLOG as inferential medium
- Aimed for general-purpose reasoning

LIMITATIONS:

- Functionality not easily generalized
- Hit same old problems beyond specific domains
- Brittle - fail unexpectedly beyond boundaries

EXAM TRAP: Expert systems = domain-specific knowledge capture!

TOPIC: Early History of Neural Networks

McCULLOCH AND PITTS (1943):

- "A logical calculus of the ideas immanent in nervous activity"
- Described activity of neurons in brain
- Proposed networks analyzed in terms of propositional logic
- Neural nets equivalent to Turing machines

DONALD HEBB (1949):

- Book: "The Organisation of Behaviour"
- Inspired by biology rather than mathematics
- Suggested associative learning in neural networks
- "Neurophysiological postulate" about connection strength

FRANK ROSENBLATT (1958):

- Developed the Perceptron model
- Rejected symbolic/algorithmic approach
- Formulated in terms of probability theory
- Inspired widespread interest in neural networks

EXAM TRAP: Early neural networks = **biologically inspired learning!**

TOPIC: Artificial Neurons

STRUCTURE:

- Takes numerical inputs (i_1, i_2, i_3 , etc.)
- Each input multiplied by weight (w_1, w_2, w_3 , etc.)
- Weights added together (plus bias term w_0)
- Net input fed into activation function f
- Output value v determined

FORMULA:

$$v = f(w_0 + w_1 i_1 + w_2 i_2 + w_3 i_3 + \dots)$$

ACTIVATION FUNCTION:

- Chosen to be non-linear
- Often approximates step function
- Triggered when net input $\geq$ threshold
- Non-linearity crucial for learning non-linear behavior

COMMON ACTIVATION FUNCTIONS:

- Sigmoid
- Hyperbolic tangent (tanh)
- Rectified Linear Unit (ReLU)

EXAM TRAP: Artificial neuron = **weighted sum** + **non-linear activation**!

TOPIC: Deep Neural Networks

STRUCTURE:

- Input layer (I)
- Multiple hidden layers (a, b, c, etc.)
- Output layer (O)
- Each neuron in one layer connected to all in next

HOW IT WORKS:

- Input layer: specific numeric activation levels represent input data
- Example: pixel color values in an image
- Hidden layers: activation depends on previous layer
- Weights vary, influencing propagation of activity
- Learning = changing weights

EXAMPLE - DIGIT RECOGNITION:

- Input: pixel values of digit image
- Output: 10 neurons (one per digit 0-9)
- Correct digit: activation close to 1
- Other digits: activation close to 0

EXAM TRAP: Deep networks = **multiple hidden layers** between input and output!

TOPIC: Minsky and Papert's Critique

THE BOOK (1969):

- "Perceptrons" by Marvin Minsky and Seymour Papert
- Fierce critique of single-layer Perceptrons
- Severely undermined enthusiasm for neural networks

KEY ARGUMENT:

- Perceptrons incapable of learning simple logical functions
- Notably anything depending on "exclusive or" (XOR)
- Appeared to wreck prospect of intelligent information processing

LATER REALIZATION:

- Critique applied only to single-layer networks
- More complex functions learnable with multiple layers
- But adding layers complicated learning process
- Route largely ignored until backpropagation in 1980s

EXAM TRAP: Minsky-Papert critique applied **only to single-layer networks!**

TOPIC: Backpropagation Learning Algorithm

PURPOSE:

- Learning algorithm for multi-layer neural networks
- Rose to prominence in 1980s
- Enabled training of deep networks

HOW IT WORKS:

- 1 Input data (e.g., image pixels)
- 2 Compare output activation to desired
- 3 Calculate how small change in output layer weights would improve match
- 4 Work back through layers (backpropagation)
- 5 Calculate for all weights in network
- 6 Adjust weights by small amount in direction of improvement
- 7 Repeat with other images (thousands/millions of times)

TECHNICAL BASIS:

- Chain rule of differentiation
- Partial derivatives of loss function
- Iterative weight adjustment

EXAM TRAP: Backpropagation = **working backward through layers** to adjust weights!

TOPIC: Connectionism

DEFINITION:

- Approach emphasizing neural networks
- Displaced pessimism of Minsky and Papert
- Demonstrated learning of complex functions

KEY WORK (1986):

- "Parallel Distributed Processing" by Rumelhart, McClelland, and PDP Research Group
- Two volumes
- Showed how layers of simple interacting neurons achieve learning

FOUR KEY ATTRACTIONS:

- 1 **Biologically Inspired:** Potential to shed light on how we think
- 2 **Natural Learning:** Through association and feedback
- 3 **General:** Applied to huge variety of tasks without special programming
- 4 **Distributed:** Robust to noisy data, generalizes well, graceful degradation

LATER DECLINE:

- Enthusiasm steeply declined in 1990s
- Failed to deliver expected practical breakthroughs
- Day would come with greater computational power

EXAM TRAP: Connectionism = **neural network approach** to cognition!

TOPIC: LeNet-5 - Convolutional Neural Network

KEY FACTS:

- Developed by Yann LeCun and colleagues (1998)
- Designed for recognition of digits
- 8 layers
- Convolutional neural network

CONVOLUTIONAL LAYERS:

- Apply local filter (kernel) repeatedly across grid
- Filter = small matrix of weights
- Weights multiplied by pixel activation values
- Sum of products = activation in convolutional layer
- Weights learned like other network weights

ARCHITECTURE:

- Input: 32×32 pixel image
- Two sequences: convolution + pooling
- Pooling reduces dimensions (averaging)
- Final layers: 120, 84, 10 neurons
- Total: 9,118 neurons

EXAM TRAP: Convolutional layers = **efficient local feature identification!**

TOPIC: AlexNet - 2012 ImageNet Breakthrough

KEY FACTS:

- Developed by Krizhevsky, Sutskever, and Hinton (2012)
- Won ImageNet Large Scale Visual Recognition Challenge
- Error rate: 15.3% (vs. 25.8% in 2011)
- Nearly 100 times bigger than LeNet-5
- Almost 900,000 neurons

ENABLING TECHNOLOGY:

- Graphics processing units (GPUs)
- Developed for animated computer games
- Process image pixels in parallel
- Recent versions: up to 10,000 cores
- Perfect for huge matrix operations in neural networks

SIGNIFICANCE:

- Startling vindication of deep networks
- Demonstrated amazing power for image recognition
- Sparked renewed interest in deep learning

EXAM TRAP: AlexNet (2012) = **breakthrough moment** for deep learning!

TOPIC: DeepMind's Atari Breakthrough

KEY FACTS (2013):

- DeepMind (London company, founded 2010)
- Deep convolutional network
- Learned to play vintage Atari games
- Including Breakout and Pong
- Performed vastly better than expert human

REMARKABLE ASPECT:

- No information about goal of games
- No information about significance of screen images
- No information about effect of user actions
- Learned entirely from trying actions
- Based on pixel information and game score

DEEP REINFORCEMENT LEARNING:

- Based on score feedback
- System teaching itself from scratch
- Learned what to do entirely on its own

EXAM TRAP: Atari program = self-taught learning from scratch!

TOPIC: AlphaGo and AlphaZero

ALPHAGO (2014-2016):

- DeepMind's deep-learning program
- Defeated European champion Fan Hui 5-0 (2014)
- Defeated World Champion Lee Sedol 4-1 (2015)
- Go considered too subtle for computer algorithms

ALPHAZERO (2017):

- More generalized program
- Capable of teaching itself new games
- Including both Go and chess
- Given: starting position, moves, rules
- Built-in: recursive adversarial tree search

KEY DIFFERENCE:

- Learned by competing against itself
- No input from human experts
- No game databases
- Self-trained in only a few hours
- Defeated world champion chess program Stockfish

EXAM TRAP: AlphaZero = self-teaching through self-play!

T26: Inscrutability of Deep Learning

TOPIC: The Inscrutability of Deep Learning

DEFINITION:

- Information stored implicitly within network weights
- Complex pattern of weights represents what network has learned
- Impossible for unaided human to interpret
- Behavior predictable only by experience

KEY CHARACTERISTICS:

- Neither weights nor roles predetermined
- Intermediate layer representations learned
- No explanation of decisions available
- "Black box" nature

ANALOGY TO HUMAN COGNITION:

- Closer to unconscious pattern-recognition
- Unlike explicit calculation or reasoning
- We also learn by examples without awareness
- Hard to articulate how we recognize things

IMPORTANT DISTINCTION:

- Machines do NOT "understand" reflectively
- Completely non-sentient
- No awareness or conscious understanding

EXAM TRAP: Inscrutability = implicit, incomprehensible representations!

T27: ChatGPT and Large Language Models

TOPIC: ChatGPT and Large Language Models

KEY FACTS (2022 onwards):

- OpenAI's chatbot
- Developed using deep learning
- Trained on 300 billion words from internet
- Potential to pass Turing Test in full generality

HOW IT WORKS:

- Statistical analysis on textual data
- Records probability that word sequences will continue
- Predicts most likely words to follow
- Chooses words (with some randomness)
- Adds word to text, repeats

TRAINING PROCESS:

- Automated learning: trillion parameters
- Supervised fine-tuning: human-crafted responses
- Reinforcement learning from human feedback
- Reward model rates responses

KEY INSIGHT:

- Method based on imitating training data
- No internal model of domain
- "Stochastic parrot"

EXAM TRAP: ChatGPT = imitating textual patterns, not understanding!

TOPIC: Limitations of ChatGPT

CHESS EXAMPLE:

- Common opening: responds with sensible moves
- Gives impression of understanding chessboard
- But: no internal model of board position
- No mechanisms of analysis or lookahead
- Subject to absurd errors:
 - Overlooking obvious captures
 - Allowing bishop to capture own piece
 - Losing track of pawns

WHY IT HAPPENS:

- Stores textual patterns, not domain knowledge
- Apparent intelligence based on illusion
- No internal understanding of what's discussed

POTENTIAL FIX:

- Link to dedicated chess engine
- Engine has model of board and analytical algorithms
- But general intelligence remains illusory

EXAM TRAP: ChatGPT's "intelligence" = *illusion based on mimicry!*

T29: Four Types of Machine Learning

TOPIC: Four Main Types of Machine Learning

1. REINFORCEMENT LEARNING:

- Sequence of actions or decisions
- Changing environment
- Trial and error
- Positive/negative feedback

2. SUPERVISED LEARNING:

- Labelled examples (training data)
- Categorize or predict
- Deep learning particularly risky
- Inscrutability and hidden biases

3. UNSUPERVISED LEARNING:

- Identify patterns without labels
- Cluster analysis
- Dimensionality reduction
- Principal Component Analysis (PCA)

4. SEMI-SUPERVISED LEARNING:

- Combination of supervised and unsupervised
- Only part of data labelled
- Unlabeled data for patterns
- Pseudo-labels

EXAM TRAP: Four types = [reinforcement](#), [supervised](#), [unsupervised](#), [semi-supervised](#)!

T30: Reinforcement Learning Applications

TOPIC: Reinforcement Learning Applications

DEFINITION:

- Sequence of actions/decisions
- Changing environment
- Goal achievement
- Trial and error with feedback

KEY APPLICATIONS:

1 Robotics:

- Teaching robots/autonomous vehicles
- Move and manipulate objects
- Interact with humans
- Learn from human feedback

2 Healthcare:

- Monitoring treatment effects
- Groups and individuals

3 Finance:

- Trading algorithms

4 Optimization:

- Any area where optimization by learning from experience

EXAMPLES FROM MODULE:

- Nim game
- Atari games
- AlphaGo
- AlphaZero

EXAM TRAP: Reinforcement learning = **learning from experience with feedback!**

T31: Supervised Learning Applications

TOPIC: Supervised Learning Applications

DEFINITION:

- Learn from labelled examples
- Categorize or predict
- Training data with labels

CLASSICAL METHODS:

- Linear regression
- Decision trees
- Support vector classification

DEEP LEARNING APPLICATIONS:

- 1 **Image Classification:**
 - Handwriting recognition
 - Face recognition
 - Road sign recognition
 - Medical scans
 - Visual scenes
- 2 **Speech Recognition**
- 3 **Machine Translation**
- 4 **Recommendation Systems**

LEARNED PREDICTION (REGRESSION):

- Numerical values rather than categories
- Financial analytics: stock prices, demand
- Asset valuation: house prices
- Credit scoring: probability of default
- Weather prediction

EXAM TRAP: Supervised learning = **learning from labelled examples!**

T32: Unsupervised Learning Applications

TOPIC: Unsupervised Learning Applications

DEFINITION:

- Identify patterns without labels
- No human input in form of labels

TWO PROMINENT FORMS:

1 Cluster Analysis:

- Groups of similar objects identified
- Without training set
- Automatic identification

2 Dimensionality Reduction:

- Main dimensions of variation identified
- Range and relationships more easily grasped
- Comparative closeness

PRINCIPAL COMPONENT ANALYSIS (PCA):

- Classical method
- Illustrates both cluster analysis and dimensionality reduction
- Application to village layout
- Application to text analysis

KEY CHARACTERISTIC:

- Less inscrutable than deep learning
- Can understand what dimensions represent
- Explore basis relatively straightforwardly

EXAM TRAP: Unsupervised learning = **finding patterns without labels!**

T33: PCA - Village Example

TOPIC: PCA - Village Layout Example

SCENARIO:

- Plot position of houses in village
- Original coordinates: latitude, longitude, height

PCA TRANSFORMATION:

- New framework aligns with best discrimination
- Village along straight road on flat land
- First vector: parallel to road (most spacing)
- Second vector: orthogonal to first
- Second vector: horizontal, distance from road
- Third vector: vertical

RESULT:

- Coordinates in new frame calculated from original
- View from above: most information
- Maps standardly do this
- Most informative presentation of layout

KEY INSIGHT:

- PCA yields what we would wish
- Dimensionality reduction
- Preserves most significant information

EXAM TRAP: PCA finds **most informative dimensions** for discrimination!

T34: PCA - Stylometry Example

TOPIC: PCA - Stylometry Example (Burrows, 1982)

QUESTION:

- Was "The Memoirs of a Lady of Quality" written by Smollett?
- Published 1751 in Smollett's Adventures of Peregrine Pickle

METHOD:

- 1 Count occurrences of all words in corpus
- 2 Order by overall frequency
- 3 Select most common 50 words
- 4 Count occurrences in each text
- 5 Divide by total word count
- 6 50 fractional values for each text

PCA APPLICATION:

- 16 texts in 50-dimensional space
- Coordinates proportional to relative frequencies
- Texts close together: similar patterns
- PCA finds best 2D projection

RESULT:

- Female authors clustered together
- "Memoirs" close to Smollett
- Fairly decisive positive verdict
- Hard to explain what similarities mean

EXAM TRAP: PCA clusters similar texts [without explicit labels!](#)

TOPIC: Risks of Inscrutability

1. HIDDEN BIAS:

- Bias in training data labels learned and perpetuated
- Presence hidden and hard to eradicate
- Example: Hiring algorithm favoring men
- Proxy variables (subjects, hobbies, careers) correlate with gender
- Removing explicit gender insufficient

2. PRIVACY RISKS:

- Complex information entangled in network
- Holds clues to personal characteristics
- Example: Facebook Likes predict sexual orientation, ethnicity, etc.
- Risks of manipulation
- Targeted advertising

3. DIFFICULTY OF ERADICATION:

- Hard to identify source of hidden biases
- Deep network provides no explanation
- True explanation hidden in vast web of weights
- Billions of weights
- Cannot be rendered humanly comprehensible

EXAM TRAP: Inscrutability = hidden bias, privacy risks, hard to eradicate!

T36: Risks of Over-Reliance

TOPIC: Risks of Over-Reliance

DEFINITION:

- Depending on system whose mode of operation seems mysterious
- Appears impressively expert
- Far faster and more confident than we could be

CONSEQUENCES:

- Undermines autonomy as responsible individuals
- Neglect to develop or apply own judgment
- Fail to realize when system is wrong

ADVERSARIAL EXAMPLES:

- Deep learning vulnerable to deliberately designed examples
- Two images differ imperceptibly to humans
- Categorized quite differently by system
- Very confidently
- Sticker on road sign causes misidentification

EXACERBATED BY HYPE:

- Excitement about ChatGPT
- Supposed general intelligence
- Software development example
- Plausible but incorrect code

EXAM TRAP: Over-reliance = automation bias - complacency and reduced vigilance!

TOPIC: Automation Bias

DEFINITION:

- Tendency of humans to over-rely on automated systems
- Leads to complacency and reduced vigilance

CONSEQUENCES:

- Errors of oversight
- Especially problematic in critical situations
- Skill atrophy
- Reduced decision-making autonomy
- Risk to safety and well-being

SKILL ATROPHY:

- Prolonged reliance on automation
- Decay in essential skills
- Operators become passive observers
- From active decision makers to passive

MITIGATION:

- Inform users of risks
- Ensure understanding of limitations
- Design mechanisms requiring periodic human input
- Keep operators engaged and alert

EXAM TRAP: Automation bias = [over-reliance leading to complacency!](#)

T38: Risks to Individuals

TOPIC: Risks to Individuals

SOURCES OF RISK:

- Organizations/institutions with whom they interact
- Their own behavior
- AI systems contributing to decision making

TYPES OF RISK:

1 Faulty Decisions:

- Poorly designed AI systems
- Trained using inappropriate data
- Biased or unfair decision making

2 Automation Bias:

- Over-reliance on automated systems
- Complacency and reduced vigilance
- Reduced decision-making autonomy
- Risk to safety and well-being

3 Manipulation:

- Algorithms drawing on large amounts of information
- Predictions about future actions
- Influence behavior
- Advertisers or interest groups

4 Privacy Threats:

- Inferences made from information
- Personal characteristics exposed

EXAM TRAP: Individual risks = **faulty decisions, automation bias, manipulation, privacy!**

TOPIC: Risks to Organizations

TYPES OF RISK:

1 Operational Risk:

- Over-reliance on AI
- False confidence
- Bad decision making
- Customer dissatisfaction
- Commercial loss

2 Reputational Risk:

- AI practices hurting individuals or groups
- Poorly designed AI algorithm
- Lack of explainability
- Privacy breaches
- Biased results

3 Regulatory Risk:

- AI-specific regulations still emerging
- Existing privacy laws apply
- Model governance regulations apply
- Potential fines and sanctions

4 Legal Risk:

- Lawsuits
- Liability
- Damages

EXAM TRAP: Organizational risks = operational, reputational, regulatory, legal!

TOPIC: Risks to Society

TYPES OF RISK:

① Economic Displacement:

- Job losses
- AI-driven applications taking place of human workers
- Widening wealth gap
- Skills gap

② Global Inequality:

- Exacerbation between countries
- Wealthier, more industrialized countries advantaged
- Less well-positioned countries disadvantaged

③ Misinformation:

- Creation and spread of misinformation
- Deep fakes
- Mislead or influence public opinion

④ Existential Risk:

- Fear of superintelligence
- Autonomous weapons systems
- Decisions against interests of creators
- No consensus on nature or extent

EXAM TRAP: Societal risks = displacement, inequality, misinformation, existential!

T41: Over-Reliance and False Confidence

TOPIC: Over-Reliance and False Confidence

THE PROBLEM:

- Apparent expertise of AI systems
- Far faster and more confident than humans
- More sophisticated processing
- More comprehensive data

CONSEQUENCES:

- Undermines autonomy
- Neglect to develop own judgment
- Fail to realize when system is wrong
- False confidence in outputs

SOFTWARE DEVELOPMENT EXAMPLE:

- ChatGPT impressive in programming exercises
- Temptation to rely on it for code production
- Economize on software engineering teams
- But: mimicking textual patterns
- Not rigorously analyzing internal model

RISKS:

- Non-standard tasks: incorrect code
- Plausible-looking code
- Works well in most cases
- Errors especially hard to identify
- False confidence

EXAM TRAP: False confidence = **plausible but incorrect outputs!**

T42: Hallucination in Large Language Models

TOPIC: Hallucination in Large Language Models

DEFINITION:

- Generate textual information completely false
- Yet often plausible
- Inevitable limitation

WHY IT HAPPENS:

- LLMs based on statistical patterns
- Not on understanding of truth
- No knowledge of the world
- Probabilistic "guesses"
- Designed for plausible responses

EXAMPLE:

- Plausible but false information
- Hard to detect
- Spread at scale
- Physical safety risks
- Legal risks

INNATE LIMITATION:

- Hallucination is inevitable
- Inherent to LLMs
- May avoid by combining with other systems
- Cannot eliminate entirely

EXAM TRAP: Hallucination = false but plausible information!

T43: Adversarial Examples

TOPIC: Adversarial Examples

DEFINITION:

- Deliberately designed inputs
- Cause deep learning models to make errors
- Often imperceptible to humans

EXAMPLE:

- Two images differ imperceptibly
- Categorized quite differently by system
- System very confident
- Sticker on road sign
- Stop sign misidentified as speed limit sign

IMPLICATIONS:

- Security vulnerability
- Safety risk
- Potential for malicious exploitation
- Challenges robustness assumptions

WHY IT MATTERS:

- Neural networks thought robust to noise
- But vulnerable to deliberate attacks
- Over-reliance especially dangerous
- Need for awareness and testing

EXAM TRAP: Adversarial examples = **imperceptible changes cause major errors!**

TOPIC: Existential Risk from AI

DEFINITION:

- Fear of "existential" risk to humanity
- Especially superintelligence
- AI system making decisions autonomously
- Against interests, values, priorities of creators or humanity

KEY CONCERNS:

- **Superintelligence:** Beyond human-level intelligence
- **AI Alignment:** Ensuring means and goals aligned with human values
- **AI Arms Race:** Competition without adequate safety measures
- **Autonomous Weapons:** Decisions without human control

NO CONSENSUS:

- Nature or extent of existential risk
- Not universally accepted concern
- Some argue current AI is "narrow"
- Far from superintelligent
- Regulatory mechanisms will develop

EXAM TRAP: Existential risk = **no consensus on nature or extent!**

T45: Narrow versus General Intelligence

TOPIC: Narrow versus General Intelligence

NARROW AI (WEAK AI):

- Designed for specific tasks
- Outperforms humans in narrow domains
- Examples: chess, image recognition, language translation
- All current practical AI

GENERAL AI (STRONG AI / AGI):

- Human-level general intelligence
- Can handle any intellectual task
- Not yet achieved
- Source of existential risk concerns

KEY DISTINCTION:

- Specific intelligence: good at one task
- General intelligence: good at many tasks
- Current AI: narrow, not general
- Turing Test: conceptual existence proof

IMPLICATIONS:

- Machines can be intelligent at specific problems
- Without general competence
- Less threatening than AGI
- Focus of practical AI applications

EXAM TRAP: Most AI is **narrow**, not general!

T46: Statistical Pattern Recognition

TOPIC: Statistical Pattern Recognition

DEFINITION:

- How neural networks work
- Based on statistical patterns in data
- Not on understanding or explicit reasoning

KEY CHARACTERISTICS:

- Learn patterns from training data
- Represent patterns implicitly in weights
- Make predictions based on patterns
- No explicit model of domain

ANALOGY TO HUMAN COGNITION:

- Similar to unconscious pattern-recognition
- Unlike explicit calculation
- We recognize things without knowing how
- Hard to articulate our own processes

IMPLICATIONS:

- Inscrutability
- Hidden biases
- Limited understanding
- "Stochastic parrot" nature

EXAM TRAP: Neural networks = [statistical pattern recognition](#), not understanding!

T47: Training Data and Bias

TOPIC: Training Data and Bias

HOW BIAS ENTERS:

- Bias in labelling of training data
- Historical prejudice
- Learned by model
- Perpetuated in outputs

HIRING EXAMPLE:

- Company historically favored men
- Data about past recruitment
- CVs and acceptance/rejection
- Model exhibits bias favoring men
- Bias assists in matching training labels

PROXY VARIABLES:

- Removing gender insufficient
- Other information acts as proxy
- Subjects at school
- Sports and hobbies
- Careers
- Strongly correlated with gender

LOAN EXAMPLE:

- Historically racially biased lending
- Segregated city areas
- Removing race from data
- Postal code correlates with race
- Perpetuates prejudice

EXAM TRAP: Removing explicit bias **insufficient** due to proxies!

T48: Privacy Risks from AI

TOPIC: Privacy Risks from AI

HOW PRIVACY IS THREATENED:

- Deep implicit entanglement of information
- Network holds clues to personal characteristics
- Characteristics we prefer to keep secret

FACEBOOK LIKES EXAMPLE:

- Used to predict sensitive attributes
- Sexual orientation
- Ethnicity
- Religious and political views
- Personality traits
- Intelligence
- Happiness
- Use of addictive substances
- Parental separation
- Age and gender

RISKS OF MANIPULATION:

- Commercial sphere
- Political sphere
- Identify personal characteristics
- Target advertising
- Influence behavior

EXAM TRAP: AI can [infer sensitive personal characteristics](#) from data!

T49: Explainable AI Challenge

TOPIC: The Explainable AI Challenge

THE PROBLEM:

- Deep network models cannot explain decisions
- True explanation hidden in vast web of weights
- Billions of weights
- Cannot be rendered humanly comprehensible

CONTRAST WITH EXPERT SYSTEMS:

- Expert systems: explicit rules
- Can explain reasoning
- Deep networks: no explanation
- No way to identify source of bias

CHALLENGE:

- Achieving explainable AI
- While retaining power of machine learning
- "Black box" to "glass box"
- Major research area

IMPLICATIONS:

- Huge reputational risk
- Regulatory challenges
- Difficulty eradicating bias
- Trust issues

EXAM TRAP: Explainable AI = [making black boxes transparent!](#)

T50: Course Navigation - Tools and Techniques

TOPIC: Course Navigation - Tools and Techniques

FOCUS:

- Mastering the AI toolbox itself
- Core building blocks of AI
- Categorization of ML algorithms

TOPICS COVERED:

- 1 **Machine Learning Algorithms:**
 - Supervised learning
 - Unsupervised learning
 - Reinforcement learning
- 2 **Data Preparation:**
 - Transforming raw information
 - Insights fit for modeling
- 3 **Feature Selection:**
 - Choosing relevant features
 - Improving model performance
- 4 **Visualization:**
 - Understanding data
 - Communicating insights
- 5 **Explainability:**
 - Balancing accuracy and explainability
 - Vital for business decision-making

KEY POINT: Mastering tools is **essential foundation!**

TOPIC: Course Navigation - Risks and Risk Factors

FOCUS:

- Potential unintended consequences of AI
- Broader perspective

TOPICS COVERED:

1 Bias Amplification:

- Existing biases amplified
- New threats to autonomy
- Safety concerns

2 Mitigation Strategies:

- Fairness
- Transparency
- Complex error translation

3 Real-World Harm:

- Errors translating into harm
- Reputational risks
- Public mistrust

4 AI Governance:

- Proactive governance
- Building trustworthy organization

KEY POINT: Understanding risks is **essential** for responsible AI!

TOPIC: Course Navigation - Ethical and Responsible AI

FOCUS:

- Moving beyond risk mitigation
- Principled implementation

TOPICS COVERED:

① Ethical Frameworks:

- Consequentialism
- Deontology
- Virtue ethics
- Applied to real-world scenarios

② Legal Landscape:

- EU AI Act
- US regulatory initiatives
- Evolving regulations

③ Proactive Approach:

- Not merely reactive
- Key leadership quality
- Rules still being shaped

KEY POINT: Ethics and regulation are **critical** for responsible AI!

TOPIC: Course Navigation - Governance

FOCUS:

- Critical lever for successful implementation
- Not bureaucratic burden

TOPICS COVERED:

1 Data Governance:

- Robust data strategy
- Ensuring data quality
- Data provenance

2 Model Governance:

- Clear processes for development
- Testing
- Ongoing monitoring

3 Outcomes:

- Reliable insights
- Avoiding unexpected liabilities
- Avoiding operational weaknesses

KEY POINT: Governance ensures **reliable and responsible AI**!

TOPIC: Course Navigation - Case Studies and Practitioner Perspectives

FOCUS:

- Bringing concepts to life
- Practical applications

TOPICS COVERED:

1 Case Studies:

- Specific applications in finance
- Risk management examples
- Real-world challenges

2 Practitioner Perspectives:

- How ML transforms risk modeling
- Practical strategies
- Ensuring explainability
- Robustness in high-stakes environments

KEY POINT: Case studies provide **practical context**!

T55: Course Goal - Business Leadership

TOPIC: Course Goal - Business Leadership

OBJECTIVE:

- Not to become a data scientist
- Rather a business leader
- Speak the language of data scientists
- Bridge gap between technical possibilities and business needs

KEY SKILLS:

- 1 Understanding AI:
 - Core concepts
 - Capabilities and limitations
- 2 Risk Management:
 - Identifying risks
 - Mitigating risks
 - Governing AI
- 3 Ethics:
 - Ethical frameworks
 - Responsible implementation
- 4 Communication:
 - Explaining AI to stakeholders
 - Translating technical to business

ULTIMATE GOAL:

- Unlock transformative potential of AI
- Sustainable
- Responsible
- Profitable

KEY POINT: Business leaders must **bridge technical and business worlds!**

TOPIC: Charles Babbage and Early Computing

KEY CONTRIBUTIONS:

- Early 19th century
- Difference engine: mechanical, gear-driven
- Could replace error-prone human calculation
- Efficient production of mathematical tables

ANALYTICAL ENGINE:

- More ambitious design
- Widely seen as anticipating digital computer
- Never completed
- Huge feat of engineering

SIGNIFICANCE:

- First practical vision of computing
- Inspired future developments
- Showed possibility of automated calculation

EXAM TRAP: Babbage = early vision of computing!

TOPIC: Alan Turing and Computability

KEY FACTS:

- 1936: Turing machine concept
- Theoretical invention
- Proved fundamental results about mathematics
- Theory of computation

TURING MACHINE:

- Universal programmable computer
- Can follow any given recipe for calculation
- Not a practical device
- Theoretical foundation

BLETCHLEY PARK:

- WWII code-breaking
- Mechanical methods of decrypting Enigma codes
- Massive impetus to electronic computer development

LATER WORK:

- 1950 paper: "Computing Machinery and Intelligence"
- Imitation game (Turing Test)
- Visionary thoughts on intelligent machines

EXAM TRAP: Turing = [founder of computability theory and AI!](#)

TOPIC: Deep Blue and Chess

KEY FACTS:

- IBM's chess-playing computer
- Defeated Garry Kasparov in 1997
- Kasparov considered strongest human player
- Conspicuous success of classical AI

CHARACTERISTICS:

- Specifically programmed to play chess
- Input from expert players
- Algorithms finely tuned to established theory
- Exhaustive search with heuristics

CONTRAST WITH ALPHAZERO:

- Deep Blue: human knowledge input
- AlphaZero: self-taught
- Deep Blue: chess-specific
- AlphaZero: generalizable

EXAM TRAP: Deep Blue = classical AI triumph!

T59: Samuel's Checkers Program

TOPIC: Samuel's Checkers Program

KEY FACTS:

- Developed by Arthur Samuel (late 1950s)
- IBM researcher
- First substantial example of machine learning
- Samuel coined the term "machine learning"

APPROACH:

- Limited exhaustive search to n moves
- Used heuristics to assess positions
- Material balance
- Mobility
- Advancement
- Kings

REINFORCEMENT LEARNING:

- Weights adjusted based on game outcomes
- Simulated evolution
- Program learned to play better than designer
- 38 criteria devised

EXAM TRAP: Samuel's checkers = [first substantial machine learning](#)!

T60: Heuristic Search - A* Algorithm

TOPIC: A* Algorithm

DEFINITION:

- Best-known heuristic search technique
- Used for route-finding
- Finds optimal path

HOW IT WORKS:

- For each location, track:
 - Distance traveled so far
 - Optimistic estimate of distance remaining
- Order locations by sum of these values
- Always explore from location with shortest sum
- Continue until destination reached

KEY REQUIREMENT:

- Estimate must be optimistic
- Cannot overestimate remaining distance
- Guarantees finding best route
- Euclidean distance suitable (crow-flies)

APPLICATIONS:

- Route planning
- Game playing
- Various optimization problems

EXAM TRAP: A* = optimal heuristic search!

T61: Nim Game - Detailed Example

TOPIC: Nim Game - Detailed Reinforcement Learning

GAME RULES:

- 20 coins on table
- Players alternate
- Remove 1, 2, or 3 coins
- Winner removes last coin(s)

WINNING STRATEGY:

- Second player has winning strategy
- Reduce to multiples of 4
- If opponent removes 1, remove 3
- If opponent removes 2, remove 2
- If opponent removes 3, remove 1
- Total of 4 removed per round

LEARNING PROCESS:

- Initial weights: all 5
- Choose randomly based on weights
- After game: adjust weights
- Winner: chosen moves +1, others -1
- Loser: chosen moves -1, others +1
- Weights range: 0 to 9

RESULT:

- After 100 games: discovers winning strategy
- Weights reflect optimal moves
- 9-0-0 pattern for multiples of 4

EXAM TRAP: Nim demonstrates [simple reinforcement learning](#)!

T62: Tower of Hanoi - Recursive Solution

TOPIC: Tower of Hanoi - Recursive Solution

PROBLEM:

- Nine disks of different sizes
- Originally piled largest to smallest
- Move entire tower from left to right
- Only one disk at a time
- Never put larger disk on smaller

RECURSIVE SOLUTION:

- To move pile of n disks from A to B:
 - 1 Move pile of $n-1$ disks from A to C
 - 2 Move disk n from A to B
 - 3 Move pile of $n-1$ disks from C to B
- Each subtask reduces to simpler instance
- Eventually reduces to single disk movements

MATHEMATICAL RESULT:

- Moving n disks takes $2^n - 1$ steps
- Moving 9 disks takes 511 steps
- Proved by mathematical induction

KEY INSIGHT:

- Complex problem solved elegantly
- No need for lookahead or comparison
- Recursion provides complete solution

EXAM TRAP: Tower of Hanoi = **elegant recursive solution!**

T63: Noughts and Crosses - Complete Algorithm

TOPIC: Noughts and Crosses - Complete Algorithm

POSITION EVALUATION:

- Value +1: Win for player to move
- Value -1: Loss for player to move
- Value 0: Draw

ALGORITHM:

- 1 If X has lost (opponent has line of three): value = -1
- 2 If position is full: value = 0
- 3 Otherwise: construct all positions with Y to move
- 4 Evaluate each from Y's point of view
- 5 Assign X-value = inverse of lowest Y-value
- 6 When turn comes, choose move with lowest opponent value

MINIMAX PRINCIPLE:

- Presume opponent chooses best reply
- Choose move minimizing opponent's maximum benefit
- Applied recursively at every level

IMPLEMENTATION:

- Fewer than 25 short lines of code
- Exhaustively analyzes all possible lines
- Creates implicit search tree

EXAM TRAP: Minimax = [adversarial search principle](#)!

T64: Heuristic Evaluation in Games

TOPIC: Heuristic Evaluation in Games

PURPOSE:

- Assess positions when exhaustive search impractical
- Guide choice of position to aim for
- Cannot guarantee optimal solution

CHECKERS CRITERIA (SAMUEL):

- 1 Material balance
- 2 Relative mobility
- 3 Advancement
- 4 Kings
- 5 38 criteria total
- 6 16 used at any time

WEIGHTED EVALUATION:

$$V = w_1 c_1 + w_2 c_2 + \dots + w_6 c_6$$

WEIGHT REFINEMENT:

- Initial weights equal
- Random variation creates competing strategy
- Play games between strategies
- Replace current if competing better
- Iterative improvement

KEY INSIGHT:

- Human judgment: identify criteria
- Reinforcement learning: optimize weights
- Program can learn to beat designer

EXAM TRAP: Heuristic weights [refined by reinforcement learning!](#)

TOPIC: Expert Systems and PROLOG

DEFINITION:

- Systems designed to capture knowledge in specific domains
- Use logic programming
- Response to limitations of general AI

PROLOG:

- Logic programming language
- Used as inferential medium
- Part of Japanese Fifth Generation project

ARCHITECTURE:

- Knowledge base: domain-specific facts
- Inference engine: general reasoning
- Knowledge engineers: extend to new domains

LIMITATIONS:

- Not easily generalized
- Same old problems beyond domains
- Brittle - fail unexpectedly
- Hard to extend

EXAM TRAP: Expert systems = domain-specific knowledge + general inference!

TOPIC: The Frame Problem

DEFINITION:

- Keeping track of what changes when action performed
- Also tracking what stays the same
- Adding to combinatorial explosion

THE CHALLENGE:

- Many aspects of situation
- Action changes some aspects
- Action doesn't change others
- Must represent both

CONSEQUENCES:

- Detailed codification adds complexity
- Fuel to combinatorial explosion
- Makes general search impractical
- Problem for classical AI

RELEVANCE:

- Shows limits of classical AI
- Need for different approaches
- Motivates machine learning

EXAM TRAP: Frame problem = tracking what changes and what doesn't!

TOPIC: Brittleness of AI Systems

DEFINITION:

- AI systems fail in unexpected ways
- Beyond familiar narrow boundaries
- Situations not explicitly foreseen

CHARACTERISTICS:

- Notorious for failing unexpectedly
- Work well within boundaries
- Fail when boundaries exceeded
- Hard to predict failures

IMPLICATIONS:

- Need for extensive testing
- Careful deployment
- Awareness of limitations
- Human oversight

RESPONSE:

- Expert systems for specific domains
- Machine learning for flexibility
- Hybrid approaches

EXAM TRAP: Brittleness = failing unexpectedly beyond boundaries!

T68: Deep Learning Renaissance

TOPIC: Deep Learning Renaissance

TIMELINE:

- 1980s: Backpropagation rises
- 1990s: Enthusiasm declines
- 1998: LeNet-5 breakthrough
- 2012: AlexNet triumph
- 2013: Atari games
- 2014-2016: AlphaGo
- 2017: AlphaZero
- 2022: ChatGPT

ENABLING FACTORS:

- Computational power (GPUs)
- Large datasets
- Algorithmic advances
- Investment and interest

KEY APPLICATIONS:

- Image recognition
- Speech recognition
- Game playing
- Language models

EXAM TRAP: Deep learning renaissance driven by **computation, data, and algorithms!**

TOPIC: ImageNet Competition

KEY FACTS:

- High-profile international competition
- Identify images from ImageNet collection
- Built by Fei Fei Li (started 2006)
- 14.2 million images
- Nearly 22,000 categories

ALEXNET WIN (2012):

- Error rate: 15.3%
- Previous year: 25.8%
- Nearly 100 times bigger than LeNet-5
- Almost 900,000 neurons

IMPACT:

- Startling vindication of deep learning
- Demonstrated amazing power
- Sparked renewed interest
- Changed computer vision field

EXAM TRAP: AlexNet (2012) = [breakthrough in image recognition!](#)

TOPIC: GPU Revolution

DEFINITION:

- Graphics Processing Units
- Developed for animated computer games
- Process image pixels in parallel

CHARACTERISTICS:

- Up to 10,000 cores
- Massive parallel processing
- Perfect for matrix operations
- Ideal for neural networks

IMPACT ON AI:

- Enabled deep learning
- Made AlexNet possible
- Accelerated training
- Democratized AI research

BROADER SIGNIFICANCE:

- Gaming industry drove innovation
- Unexpected benefit for AI
- Example of technology transfer

EXAM TRAP: GPUs enabled **deep learning revolution!**

T71: AlphaGo - Significance

TOPIC: AlphaGo - Significance

GO GAME:

- Ancient Chinese board game
- More complex than chess
- Average branching factor 250
- Considered too subtle for computers

ALPHAGO ACHIEVEMENTS:

- Defeated Fan Hui 5-0 (2014)
- Defeated Lee Sedol 4-1 (2015)
- World champion defeated
- Historic milestone

TECHNIQUES:

- Deep convolutional neural networks
- Reinforcement learning
- Monte Carlo tree search
- Combination of approaches

SIGNIFICANCE:

- Showed AI can master complex games
- Beyond human capability
- Not specifically programmed
- Learned from experience

EXAM TRAP: AlphaGo = **mastering complex game of Go!**

T72: AlphaZero - Generalization

TOPIC: AlphaZero - Generalization

ADVANCE OVER ALPHAGO:

- More generalized program
- Capable of teaching itself new games
- Go and chess
- Not just one game

WHAT IT WAS GIVEN:

- Starting position
- Moves of all pieces
- Various rules
- Win, draw, loss identification
- Efficient recursive adversarial tree search

HOW IT LEARNED:

- Competing against itself
- No human expert input
- No game databases
- Self-training for few hours
- Defeated Stockfish (world champion chess program)

SIGNIFICANCE:

- Massive progress toward AGI
- Self-taught skills
- Beyond human ingenuity
- Potentially frightening

EXAM TRAP: AlphaZero = [self-teaching through self-play](#)!

T73: Stochastic Parrot

TOPIC: Stochastic Parrot

DEFINITION:

- Term coined by Emily M. Bender (2021)
- Describes LLMs like ChatGPT
- Generates plausible text without understanding
- Based on statistical patterns

CHARACTERISTICS:

- Imitates responses from training data
- With variation
- No internal model of domain
- No understanding of what's discussed

EVIDENCE:

- Chess example
- Common openings: sensible moves
- Uncommon openings: errors appear
- No board model
- No analysis or lookahead

IMPLICATIONS:

- Apparent intelligence is illusion
- Cannot trust outputs
- Need for verification
- Hallucination inevitable

EXAM TRAP: Stochastic parrot = [mimicking without understanding!](#)

T74: Reinforcement Learning from Human Feedback

TOPIC: Reinforcement Learning from Human Feedback (RLHF)

DEFINITION:

- Training technique for LLMs
- Humans assess relative suitability of responses
- Feedback used to generate reward model

PROCESS:

- 1 System generates range of responses for prompt
- 2 Humans assess relative suitability
- 3 Statistical analysis of feedback
- 4 Generate "reward model"
- 5 Use reward model to rate responses
- 6 Select most appropriate responses

APPLICATION:

- ChatGPT training
- Improving response quality
- Aligning with human preferences
- Reducing harmful outputs

LIMITATIONS:

- Based on human judgments
- Humans may have biases
- Cannot eliminate all issues
- Hallucination remains inevitable

EXAM TRAP: RLHF = using human feedback to improve LLMs!

T75: Supervised Fine-Tuning

TOPIC: Supervised Fine-Tuning

DEFINITION:

- Training technique for LLMs
- Human-crafted responses to typical prompts
- Supervised learning approach

PROCESS:

- 1 Collect typical prompts
- 2 Human experts craft suitable responses
- 3 Train model on these examples
- 4 Model learns to generate similar responses

PURPOSE:

- Improve response quality
- Align with human expectations
- Reduce inappropriate outputs
- Guide model behavior

RELATIONSHIP TO RLHF:

- First stage of training
- Followed by RLHF
- Both use human input
- Complementary approaches

EXAM TRAP: Fine-tuning = **learning from human-crafted examples!**

T76: Principal Component Analysis - Definition

TOPIC: Principal Component Analysis (PCA) - Definition

DEFINITION:

- Classical method for dimensionality reduction
- Identifies main dimensions of variation
- Enables easier grasp of range and relationships

HOW IT WORKS:

- Transform coordinates
- New dimensions align with best discrimination
- First vector: most discriminating
- Second vector: orthogonal, next most
- Continue for all dimensions

APPLICATIONS:

- Village layout example
- Stylometry example
- Data visualization
- Feature extraction

ADVANTAGES:

- Powerful dimensionality reduction
- Preserves significant information
- Highlights clusters
- Less inscrutable than deep learning

EXAM TRAP: PCA = finding most informative dimensions!

TOPIC: Dimensionality Reduction

DEFINITION:

- Technique in unsupervised learning
- Identifies main dimensions of variation
- Enables easier grasp of range and relationships

PURPOSE:

- Simplify complex data
- Reduce computational requirements
- Visualize high-dimensional data
- Remove redundant features

METHODS:

- Principal Component Analysis (PCA)
- Autoencoders
- t-SNE
- UMAP

PCA ADVANTAGE:

- Less inscrutable than deep learning
- Understand what dimensions represent
- Explore basis straightforwardly

EXAM TRAP: Dimensionality reduction = [simplifying complex data](#)!

T78: Cluster Analysis

TOPIC: Cluster Analysis

DEFINITION:

- Learning technique in unsupervised learning
- Groups of similar objects identified
- Without training set
- Automatic identification

PURPOSE:

- Discover natural groupings
- Segment data
- Identify patterns
- Exploratory analysis

METHODS:

- K-means clustering
- Hierarchical clustering
- DBSCAN
- Gaussian mixture models

APPLICATIONS:

- Customer segmentation
- Document clustering
- Image segmentation
- Anomaly detection

EXAM TRAP: Cluster analysis = automatic grouping of similar objects!

T79: Data Leakage

TOPIC: Data Leakage

DEFINITION:

- Using future or concurrent data
- Wouldn't have been observable at training time
- Causes optimistic performance estimates

EXAMPLE:

- Predicting loan defaults
- Using data from after loan approval
- Model appears accurate
- But fails in production

PREVENTION:

- Careful data splitting
- Time-based validation
- Cross-validation
- Domain knowledge

CONSEQUENCES:

- Overly optimistic performance
- Poor generalization
- Failed deployments
- Financial losses

EXAM TRAP: Data leakage = using future data in training!

TOPIC: Model Robustness

DEFINITION:

- Measure of how well model performs
- With inputs previously not encountered
- Or changes in data

IMPORTANCE:

- Real-world data varies
- Conditions change
- Edge cases exist
- Adversarial attacks

CHALLENGES:

- Deep learning vulnerable to adversarial examples
- Small changes cause major errors
- Overconfidence in wrong predictions
- Difficulty detecting failures

IMPROVEMENT:

- Extensive testing
- Adversarial training
- Ensemble methods
- Regularization

EXAM TRAP: Robustness = performing well on unseen data!

TOPIC: Model Interpretability

DEFINITION:

- Degree to which human can comprehend model behavior
- With built-in mechanisms
- Understand how inputs affect outputs

SPECTRUM:

- **High:** Decision trees, linear regression
- **Medium:** Random forests, gradient boosting
- **Low:** Deep neural networks

TRADE-OFF:

- Complex models often more accurate
- But less interpretable
- Need to balance
- Depends on application

TOOLS:

- Feature importance
- SHAP values
- LIME
- Counterfactual explanations

EXAM TRAP: Interpretability vs. accuracy **trade-off!**

TOPIC: Model Explainability

DEFINITION:

- Capacity to explain how AI makes decisions
- Often after the fact (ex-post)
- Understandable terms

IMPORTANCE:

- Accountability
- Trust building
- Error identification
- Bias detection

CHALLENGES:

- Deep networks opaque
- Billions of weights
- No simple explanation
- Hard to identify bias sources

APPROACHES:

- Local explanations
- Global explanations
- Model-specific methods
- Model-agnostic methods

EXAM TRAP: Explainability = explaining decisions after the fact!

T83: Training Data Quality

TOPIC: Training Data Quality

IMPORTANCE:

- Models only as good as training data
- Garbage in, garbage out
- Quality determines performance

KEY ATTRIBUTES:

- Accuracy
- Completeness
- Consistency
- Relevance
- Timeliness

COMMON ISSUES:

- Missing values
- Outliers
- Noisy data
- Biased sampling
- Label errors

BEST PRACTICES:

- Data validation
- Preprocessing
- Documentation
- Continuous monitoring

EXAM TRAP: Training data quality **determines model quality!**

T84: Feature Engineering

TOPIC: Feature Engineering

DEFINITION:

- Selecting and engineering appropriate features
- Variables used as inputs for model
- Critical for model performance

ACTIVITIES:

- Transforming variables
- Combining variables
- Creating derived features
- Dimensionality reduction

EXAMPLES:

- Log transformations
- Polynomial features
- Interaction terms
- Domain-specific ratios

IMPORTANCE:

- Good features improve performance
- Reduces need for complex models
- Domain knowledge valuable
- Iterative process

EXAM TRAP: Feature engineering = [creating informative inputs!](#)

T85: Model Selection

TOPIC: Model Selection

DEFINITION:

- Choosing modeling approach
- Comprehensive evaluation of alternatives
- Documenting rationale

CONSIDERATIONS:

- Problem type (classification, regression)
- Data characteristics
- Interpretability requirements
- Computational resources
- Regulatory requirements

APPROACHES:

- Start simple
- Compare alternatives
- Cross-validation
- Consider trade-offs

DOCUMENTATION:

- Rationale for selection
- Alternatives considered
- Performance comparison
- Limitations

EXAM TRAP: Model selection requires **systematic evaluation**!

T86: Hyperparameter Tuning

TOPIC: Hyperparameter Tuning

DEFINITION:

- Optimizing parameters not learned from data
- Set before training
- Configuration settings

EXAMPLES:

- Learning rate
- Number of hidden layers
- Regularization parameter
- Number of epochs

METHODS:

- Grid search
- Random search
- Bayesian optimization
- Cross-validation

IMPORTANCE:

- Affects model performance
- Balance bias-variance
- Avoid overfitting
- Optimize generalization

EXAM TRAP: Hyperparameters = [settings chosen before training!](#)

T87: Cross-Validation

TOPIC: Cross-Validation

DEFINITION:

- Technique for model evaluation
- Split data into folds
- Train on some, validate on others
- Repeat and average

K-FOLD CROSS-VALIDATION:

- Split data into k folds
- Train on k-1 folds
- Validate on 1 fold
- Repeat k times
- Average results

BENEFITS:

- Efficient use of data
- Reliable performance estimate
- Reduces overfitting
- Model selection

VARIANTS:

- Stratified (preserves class distribution)
- Leave-one-out (LOOCV)
- Time series (preserves order)

EXAM TRAP: Cross-validation = [train on k-1, validate on 1!](#)

TOPIC: Model Testing Types

CHRONOLOGICAL ORDER:

- 1 **Unit Tests:** Individual code pieces
- 2 **Component Tests:** Individual code sections
- 3 **Integration Tests:** Interfaces between components
- 4 **System Tests:** Complete integrated system
- 5 **Performance Tests:** Speed and stability
- 6 **Regression Tests:** After modifications
- 7 **Acceptance Tests:** Ready for deployment
- 8 **Adversarial Tests:** Extreme scenarios

TESTING APPROACHES:

- White-box: access to code
- Black-box: user perspective
- Gray-box: knowledge of architecture

KEY PRINCIPLES:

- Start testing early
- Use realistic configurations
- Less realistic parameters better
- Continuous testing

EXAM TRAP: Testing should be **comprehensive and start early!**

T89: White-Box Testing

TOPIC: White-Box Testing

DEFINITION:

- Testers have access to internal data structures
- Access to algorithms
- Access to actual code

CHARACTERISTICS:

- Testing at unit level
- From developer's perspective
- Line-by-line proofreading
- More effective for finding specific issues

WHEN TO USE:

- Internally developed models
- Early stage of development
- When code access available

ADVANTAGES:

- Thorough testing
- Can identify specific issues
- Deeper understanding

DISADVANTAGES:

- May introduce bias
- Limited to accessible code
- Not suitable for proprietary models

EXAM TRAP: White-box = [access to internal code!](#)

T90: Black-Box Testing

TOPIC: Black-Box Testing

DEFINITION:

- Treats software as closed box
- From user's perspective
- No knowledge of internal implementation

CHARACTERISTICS:

- Reduces likelihood of bias
- Functional testing
- Focus on inputs and outputs
- User experience testing

WHEN TO USE:

- Proprietary vendor models
- Functional testing
- Final stages of development
- When code not accessible

ADVANTAGES:

- Reduces bias
- Tests actual user experience
- Suitable for all models

DISADVANTAGES:

- Less thorough
- Harder to identify specific issues
- Cannot see internal workings

EXAM TRAP: Black-box = user perspective, no code access!

T91: Gray-Box Testing

TOPIC: Gray-Box Testing

DEFINITION:

- Between white-box and black-box
- Informed by knowledge of functional specifications
- Knowledge of architecture
- Does not access code itself

CHARACTERISTICS:

- Combines benefits of both approaches
- Knowledge of system design
- No direct code access
- Balanced perspective

WHEN TO USE:

- Distributed applications
- Testing component integration
- When architecture known
- Complex systems

ADVANTAGES:

- More informed than black-box
- Less biased than white-box
- Flexible approach

EXAM TRAP: Gray-box = knowledge of architecture, not code!

TOPIC: The Use Test

DEFINITION:

- Examines model within context
- Considers human interaction
- Assesses actual usage
- Evaluates acceptance

KEY QUESTIONS:

- Are results utilized appropriately?
- Do they prove useful?
- If not, why?

FOCUS AREAS:

- People, not numbers
- Acceptance and trust
- Comprehensibility
- Communication

REPORTING:

- Presented to senior management
- Not technical reports
- Qualitative in nature
- Improves culture

EXAM TRAP: Use test focuses on [how people use the model!](#)

T93: Model Limitations

TOPIC: Model Limitations

DEFINITION:

- Constraints on model applicability
- Boundaries beyond which model invalid
- Should be documented
- Must be communicated to users

TYPES:

- **Product Applicability:** Specific products only
- **Access Restrictions:** Who can use
- **Portfolio Size Limits:** For new markets
- **Data Limitations:** Quality, coverage

WHY LIMITS:

- Prevent misuse
- Ensure appropriate application
- Maintain risk ranking
- Manage expectations

DOCUMENTATION:

- In model inventory
- In model documentation
- Communicated to users
- Regular review

EXAM TRAP: Model limits prevent misuse beyond intended application!

TOPIC: Model Risk Ranking

DEFINITION:

- Classification of model risk
- Drives validation depth and frequency
- Based on multiple factors

KEY FACTORS:

1 Materiality:

- Economic consequences
- Operational impact
- Reputational consequences

2 Complexity:

- May elevate ranking
- Even if materiality low

3 Exposure:

- Published financial reports
- Wide impact

IMPORTANT:

- Minimum standards for all models
- Regardless of materiality
- Absence of assessment is risk

EXAM TRAP: Risk ranking considers **materiality**, **complexity**, **exposure**!

T95: Three Lines of Defense

TOPIC: Three Lines of Defense

FIRST LINE:

- Business and model owners/developers
- Own the model and performance
- Design and implement models
- First to react to breaches

SECOND LINE:

- Independent risk management/validation
- Set policy
- Conduct validations
- Challenge First Line

THIRD LINE:

- Internal audit
- Independent assurance
- Review framework effectiveness
- Ensure compliance

COMMUNICATION:

- Breach notification
- Independent review
- Dispute resolution
- Reporting

EXAM TRAP: Three lines provide [checks and balances](#)!

T96: Model Validation

TOPIC: Model Validation

DEFINITION:

- Rigorously testing model, inputs, outputs, governance
- Identify extent of residual model risk
- Assess mitigating controls

WHEN PERFORMED:

- **Initial:** Before implementation
- **Periodic:** Annual or more frequent
- **Change-Based:** After material changes

KEY ACTIVITIES:

- Data assessment
- Assumptions review
- Limitations identification
- Implementation review
- Performance monitoring
- Documentation review

KEY PRINCIPLE:

- No "valid" model
- Attempt to invalidate
- Usefulness vs. validity
- Effective challenge

EXAM TRAP: Validation = attempt to invalidate, not prove validity!

T97: Model Changes

TOPIC: Model Changes

CHANGE MANAGEMENT:

- Structured process
- Change requests
- Impact assessment
- Documentation
- Validation
- Approval

INDEPENDENT REVIEW:

- By individuals not involved
- Assess accuracy, reliability
- Compliance
- Potential risks

AI/ML DIFFERENCES:

- More frequent monitoring
- Broader stakeholders
- Focus on explainability
- Benchmarking
- Different data sets
- Specific cases

IMPLEMENTATION:

- Controlled manner
- Documentation
- Monitoring changes
- Automated controls

EXAM TRAP: Model changes require [structured governance](#)!

T98: Data Governance

TOPIC: Data Governance

DEFINITION:

- System of decision rights and accountabilities
- For information-related processes
- People, processes, technologies

KEY COMPONENTS:

- 1 **Strategy:** Vision, goals, policies
- 2 **Quality:** Accuracy, consistency, integrity
- 3 **Provenance:** Legal right to use
- 4 **Classification:** Structure, confidentiality
- 5 **Metadata:** Documentation, dictionary
- 6 **Security:** Protection, compliance

THREE PRINCIPLES:

- Obtaining consent
- Minimizing data held
- Ensuring rights of data subjects

BOARD ROLE:

- Approve framework
- Oversee compliance
- Assess risks
- Regular review

EXAM TRAP: Data governance = **decision rights and accountabilities!**

TOPIC: GenAI Governance

WHY HARDER:

- Opacity and unpredictability
- Emergent capabilities
- Post-deployment shifts
- Embedded in workflows
- Fragmented regulation

INTERNAL LEVERS:

- Assign ownership
- Risk-based review
- Incentivize responsible behavior
- Staff training
- Escalation paths

EXTERNAL CONSTRAINTS:

- Track regulations
- Evaluate provider responsibilities
- Align with standards

ADAPTIVE TOOLS:

- Continuous monitoring
- Red-teaming
- Scenario planning
- Technical safeguards
- Feedback loops

EXAM TRAP: GenAI governance requires [internal](#), [external](#), [adaptive approaches](#)!

T100: AI and Risk - Final Summary

TOPIC: AI and Risk - Comprehensive Summary

CLASSICAL AI:

- Explicit, rule-based processing
- Search, recursion, lookahead
- Limits: brittleness, frame problem

NEURAL NETWORKS:

- Biologically inspired
- Deep learning, backpropagation
- Inscrutability challenges

MACHINE LEARNING TYPES:

- Reinforcement, supervised, unsupervised, semi-supervised

RISKS:

- Inscrutability: hidden bias, privacy
- Over-reliance: automation bias
- Individual, organizational, societal

KEY TAKEAWAYS:

- AI is powerful but risky
- Understanding limitations crucial
- Responsible governance essential
- Balance innovation with risk management

EXAM SUCCESS: Understand [classical AI](#), [neural networks](#), [ML types](#), and [AI risks](#)!